

HOSAM ALKHAWALDEH

AI Engineer | Electrical Power Engineering

Amman, Jordan • +962 79 064 7724 • hosam70alkhwalda@gmail.com

[Portfolio](#) • [GitHub](#) • [LinkedIn](#)

PROFESSIONAL SUMMARY

AI Engineer with an Electrical Power Engineering foundation, building applied NLP, retrieval-augmented generation (RAG), and knowledge-graph systems for bilingual Arabic/English use cases. Ships production-oriented services end to end: FastAPI microservices, containerized deployment with Docker Compose, Prometheus-based observability, and automated evaluation harnesses rather than one-off notebooks.

Recent builds include a multi-service RAG and knowledge-graph platform with deterministic NL-to-Cypher translation, and a bilingual voice-based AI interviewer with speech recognition, text-to-speech, and native Arabic PDF reporting. Brings the reliability standards of power-systems engineering to AI system design: measurable evaluation, resilient service architecture, and clean deployment practices.

Core focus: applied NLP, RAG architectures, and Arabic-language AI systems — built to a standard where they can be evaluated, monitored, and deployed, not just prototyped.

KEY AI PROJECTS

Multi-Service RAG & Knowledge-Graph Platform — Python, FastAPI, Neo4j, Docker Compose, Prometheus

- Designed a three-service architecture (NER-KG, RAG, Router) with a classify → fan-out → aggregate pipeline and health-checked service dependencies.
- Built a deterministic NL-to-Cypher translation layer over a Neo4j knowledge graph, plus a GraphRAG hybrid retrieval pipeline combining symbolic and vector search.
- Implemented an automated evaluation harness scoring NER F1 = 0.87, Cypher-translation accuracy = 91%, and RAG grounding rate = 88%, backed by Prometheus metrics and ASGI-level observability middleware.

Adversarial NLP Evaluation & Domain-Shift Analysis — Python, PyTorch, Hugging Face Transformers

- Fine-tuned DistilBERT for sentiment classification, then built adversarial QA evaluation datasets specifically designed to expose positional bias in model predictions.
- Ran domain-shift analysis on 1,033 out-of-domain news articles: mean predicted confidence dropped to 0.628, with 50.5% of predictions falling below the 0.60 threshold — quantifying systematic degradation on crime- and misconduct-related content.
- Focused on model robustness and evaluation rigor over raw accuracy — surfacing failure modes before deployment rather than after.

Hybrid Search & Re-Ranking Pipeline — Python, Weaviate, sentence-transformers

- Layered a cross-encoder re-ranking stage on top of a Weaviate hybrid (keyword + vector) search pipeline, improving Recall@10 from 71% to 85% and documenting the evaluation results.

More projects and code: [Portfolio](#) / [GitHub](#)

SKILLS

- AI / ML: Python, PyTorch, scikit-learn, NumPy, Pandas, RAG pipelines, LLM fine-tuning, LLM agents
- NLP: Hugging Face Transformers, spaCy, BERT, RoBERTa, sentence-transformers, embeddings
- Deployment & Observability: FastAPI, Docker, Docker Compose, Prometheus, REST APIs, Git, GitHub, CI workflows
- Databases & Knowledge Graphs: PostgreSQL, SQLite, Weaviate, Neo4j, SPARQL, ETL pipelines
- Electrical Engineering: Power distribution systems, low-voltage engineering, transformer & cable testing, SCADA
- Programming Languages: Python, C++, SQL

PROFESSIONAL EXPERIENCE

Electrical Engineer / Manager

Roya Design

Aug 2024 – June 2026

Amman, Jordan

- Designed and directly supervised electrical plans across 15+ interior design and decoration projects in Amman.
- Maintained full compliance with safety and quality standards across all active sites, with zero recorded violations.
- Coordinated with architects, contractors, and site teams, delivering all projects on schedule and to specification.

Trainee Engineer

Mar 2024 – Jul 2024

EDCO — Electricity Distribution Company

Jordan

- Implemented and supervised electrical specifications aligned with IEC/IEEE standards across all 4 departments — Specifications, Protection, Renewable Energy, and SCADA.
- Analyzed electrical faults on the distribution network and developed preventive solutions that reduced recurring fault incidents.
- Conducted transformer examinations and cable-factory conformance testing, confirming compliance on every unit inspected.

Instructor & Course Developer

Feb 2023 – Feb 2024

Electric Youth Committee — Jordanian Engineers Association

Jordan

- Designed and delivered electronics and electrical engineering course material combining theory with hands-on practice, training 20+ aspiring engineers.

APPLIED AI TRAINING

AI.SPIRE — Applied NLP & Knowledge Systems, LevelUp Economy / Istidama Consulting (Feb 2026 – July 2026). 24-week program covering ML pipelines, NLP fine-tuning, RAG, knowledge graphs, and production deployment — the foundation behind the projects above.

EDUCATION

AI.SPIRE: Applied AI and ML Systems

2026

LevelUp Economy & Istidama Consulting

Jordan

Python for AI — Applied NLP & Knowledge Systems

- 24-week intensive program covering applied NLP, machine learning, RAG systems, and Knowledge Graphs.
- Built and deployed end-to-end AI systems integrating NLP pipelines, vector search, and APIs.
- Developed professional skills in technical communication, teamwork, and AI system documentation.

BSc in Electrical Power Engineering

2020 – 2024

Al-Balqa Applied University — Faculty of Engineering Technology

Jordan

- Graduation Project: Photovoltaic-Thermal (PVT) System — improved solar panel efficiency by cooling panels and using excess heat for steam generation.

COURSES & CERTIFICATIONS

- Synchronous Generator Systems — Committee of Engineers in the Electricity Sector, 2024

LANGUAGES

- Arabic — Native
- English — Professional Working Proficiency